AN ASCENDING CHAIN OF SUBSHEAVES OF A CONSTRUCTIBLE SHEAF ON A NOETHERIAN SCHEME STABILISES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the inheritance of constructibility from the Noetherian proposition and the stabilisation from the finite generation property.

Editorial note. The label lemma-constructible-over-noetherian-noetherian names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Constructibility over a Noetherian scheme is inherited by subsheaves and colimits. Same chapter, the proposition stating that on a Noetherian scheme the constructible sheaves of sets, abelian groups or modules form a well-behaved family closed under the relevant operations. Supplies that the members of the chain and their colimit are constructible.
- Constructible sheaves are finitely generated under filtered colimits. Same chapter, the lemma expressing a constructible sheaf as the union of the images of maps from finitely many generators. Supplies the stabilisation of the chain.

Source reference. Original reference 095P: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11903–11903. Prerequisite pointer invoked at proof line(s) 12917; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09BH: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 12804–12804. Prerequisite pointer invoked at proof line(s) 12914; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a Noetherian scheme. Let Λ be a Noetherian ring. Consider inclusions*

$$\mathcal{F}_1 \subset \mathcal{F}_2 \subset \mathcal{F}_3 \subset \ldots \subset \mathcal{F}$$

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in the category of sheaves of sets, abelian groups, or Λ-modules. If $\mathcal{F}$ is constructible, then for some n we have $\mathcal{F}_n = \mathcal{F}_{n+1} = \mathcal{F}_{n+2} = \ldots$.

Proof. By Proposition 09BH we see that $\mathcal{F}_i$ and $\operatorname{colim} \mathcal{F}_i$ are constructible. Then the lemma follows from Lemma 095P. □